MILITARY SPECIFICATION SHEET

RELAYS, HYBRID, SENSORS, ESTABLISHED RELIABILITY,
TYPE IV (UNDER/OVER FREQUENCY SENSOR), CLASS B, 4PDT,
LOW LEVEL TO 10 AMPERES, HERMETICALLY SEALED (POTTED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-28894.

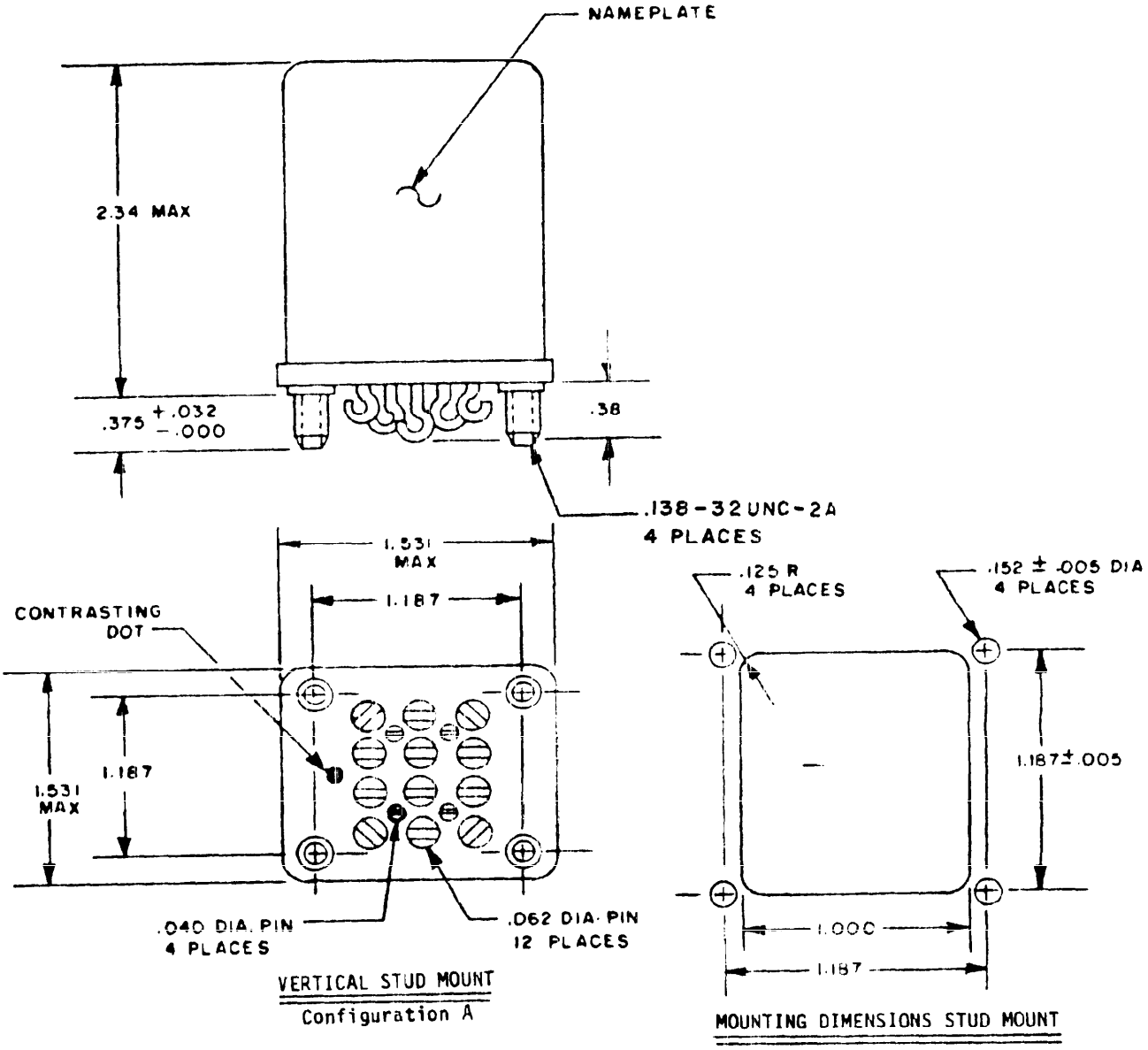
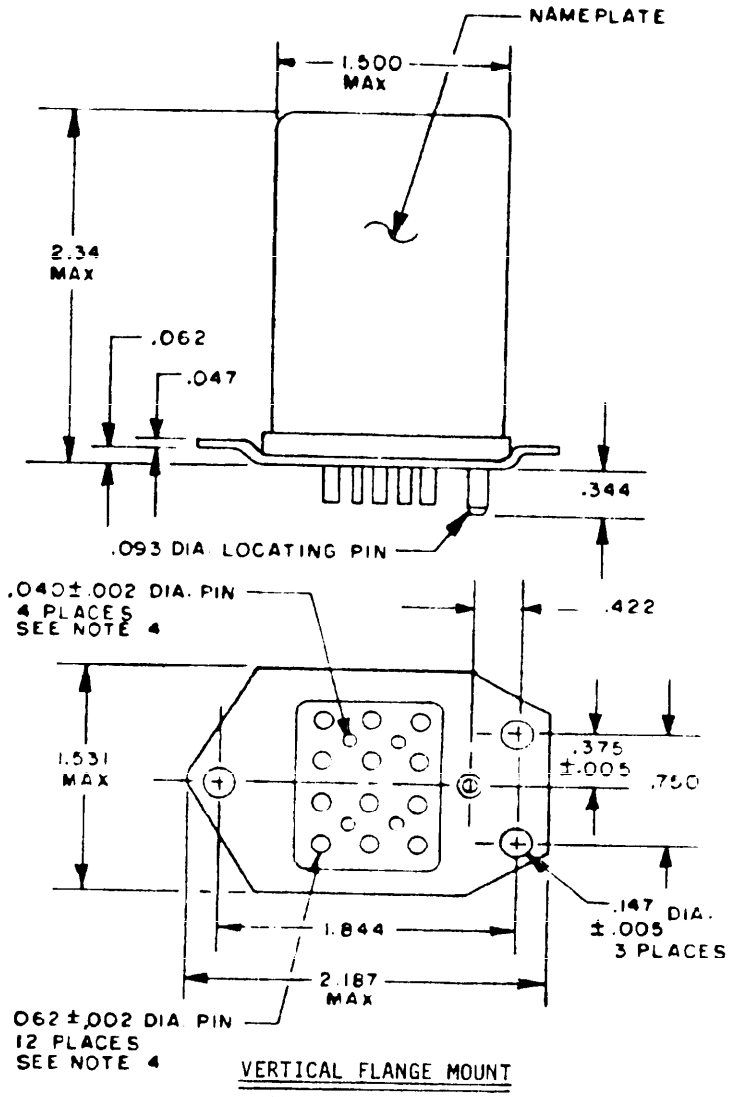


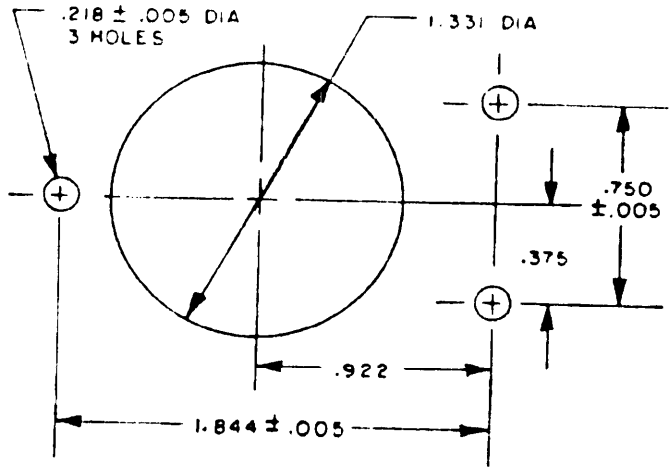

FIGURE 1. Dimensions and configuration.


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VERTICAL FLANGE MOUNT

Configuration B

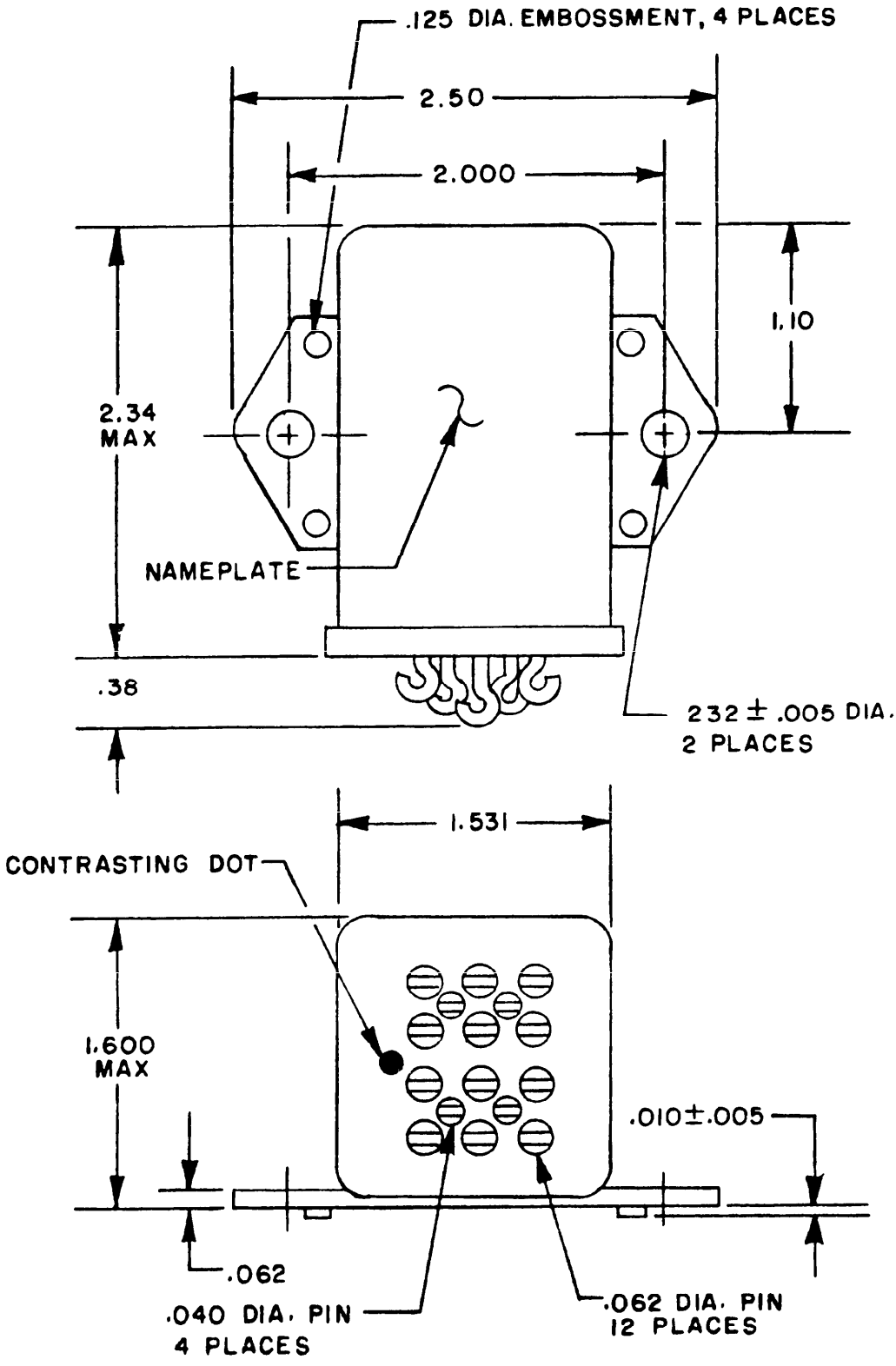



MOUNTING DIMENSIONS FLANGE MOUNT

FIGURE 1. Dimensions and configuration - Continued.



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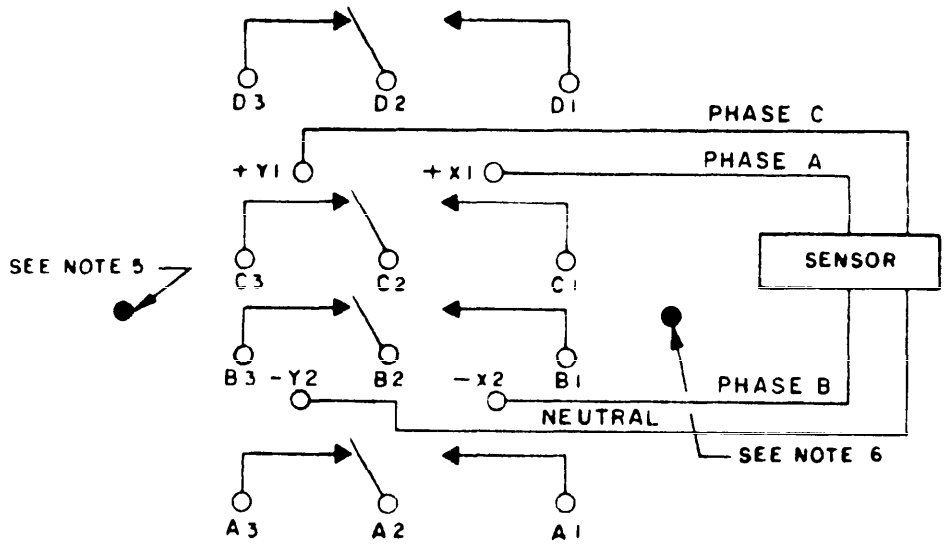



HORIZONTAL FLANGE MOUNT
Configuration C

FIGURE 1. Dimensions and configuration - Continued.



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CIRCUIT DIAGRAM TERMINAL VIEW

| Pin | Input Connection |
|---|---|
| +X1 | Phase A |
| -X2 | Phase B |
| +Y1 | Phase C |
| -Y2 | Neutral |

| Inches | mm | Inches | mm |
|---|---|---|---|
| .000 | 0.00 | .375 | 9.53 |
| .002 | 0.05 | .38 | 9.7 |
| .005 | 0.13 | .750 | 19.05 |
| .010 | 0.25 | .922 | 23.42 |
| .032 | 0.81 | 1.000 | 25.40 |
| .040 | 1.02 | 1.10 | 27.9 |
| .062 | 1.58 | 1.187 | 30.15 |
| .093 | 2.36 | 1.500 | 38.10 |
| .125 | 3.18 | 1.531 | 38.89 |
| .138 | 3.51 | 1.844 | 46.83 |
| .147 | 3.73 | 1.887 | 47.93 |
| .152 | 3.86 | 2.187 | 55.55 |
| .218 | 5.54 | 2.34 | 59.4 |
| .344 | 8.74 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Terminal numbers shown above for reference only. Numbers do not appear on header.
4. Finish shall be gold plate per MIL-G-45204, type II, class 1, underplating shall be nickel 50 to 150 microinches.
5. A contrasting dot shall appear on header and circuiting diagram for vertical stud mount and horizontal flange mount only.
6. A locating pin designator shall appear on circuit diagram for vertical flange mount only.
7. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.

FIGURE 1. Dimensions and configuration - Continued.



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REQUIREMENTS:

Input data:

Input voltage range: 90 V ac to 124 V ac, line to neutral. 10% maximum harmonic distortion.

Input frequency range: 40 Hz to 480 Hz.

Under frequency: 320 Hz.

Over frequency: 480 Hz.

Sensing point accuracy: ±2% maximum.

Operating mode: The output relay energizes when the frequency is above the specified limit and de-energizes when the frequency is below the specified limit.

Operating data:

Pickup time (electromechanical relay): 50 milliseconds maximum when sensed voltage is at or above the trip point.

Dropout time (electromechanical relay): 50 milliseconds maximum when sensed voltage is below the trip point.

Current: Maximum operating current of 75 milliamperes.

Contact data:

Configuration: 4PDT, 10 amperes, integral class B relay.

Load ratings:

High level (relay case grounded):

Resistive:
10 amperes at 28 V dc, 115 V ac 400 Hz and 115/200 V ac 400 Hz.
2.5 amperes at 115 V ac 50/60 Hz and 115/200 V ac 50/60 Hz. 1/

Inductive:
8 amperes at 28 V dc, 115 V ac 400 Hz and 115/200 V ac 400 Hz (20,000 cycles minimum life).
2.5 amperes at 115 V ac 50/60 Hz and 115/200 V ac 50/60 Hz. 1/

Motor:
4 amperes at 28 V dc, 115 V ac 400 Hz and 115/200 V ac 400 Hz.
2 amperes at 115 V ac 50/60 Hz and 115/200 V ac 50/60 Hz. 1/

Lamp:
2 amperes at 28 V dc, 115 V ac 400 Hz, and 115/200 V ac 400 Hz.
1 ampere at 115 V ac 50/60 Hz and 115V/200 V ac 50/60 Hz. 1/

Low level: 10 to 50 microamperes at 10 to 50 millivolts (dc or peak ac).

Intermediate current: Applicable.

Contact voltage drop:

High level:

Before life: 0.150 volt maximum.

After life: 0.175 volt maximum.

1/ For 50/60 Hz, overload is not applicable and life shall be 10,000 cycles.



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slification of MIL-R-28894/15 relays may be performed reduced testing for qu

Low level:

Before life: 0.05 ohm maximum.

After life: 0.15 ohm maximum.

Contact bounce: 1.0 millisecond maximum (applicable to failure rate level "L").

Contact stabilization time: 1.5 milliseconds maximum (applicable to failure rate levels "M", "P", and "R").

Overload: 40 amperes dc, 60 amperes ac.

Electrical data:

Insulation resistance: 100 megohms minimum initial; 50 megohms after environmental or life tests.

Dielectric withstanding voltage:

| | Initial | After life tests |
|---|---|---|
| Input to case: | 1000 V rms (60 Hz) | 1000 V rms (60 Hz) |
| All other points: | 1250 V rms (60 Hz) | 1000 V rms (60 Hz) |

| | Altitude | |
|---|---|---|
| | 80,000 ft | 300,000 ft |
| Input to case: | 350 V rms (60 Hz) | 500 V rms (60 Hz) |
| All other points: | 350 V rms (60 Hz) | 500 V rms (60 Hz) |

Voltage trip points sensing range: 1 V dc to 50 V dc.

Voltage transients: MIL-STD-704, CAT-B.

Timing cycle: Not applicable.

Environmental data:

Temperature range: -55°C to +125°C.

Thermal shock: Applicable.

Vibration: MIL-STD-202, method 204, test condition D, except frequency shall be 5 to 2,000 Hz.

Vibration, random: MIL-STD-202, method 214, test condition IG, 15 minutes each plane.

Shock: MIL-STD-202, method 213, test condition A (50G's).

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Physical data:

Seal: Hermetic, potted.

Dimensions and configuration: See figure 1.

Terminal strength: Method 211 of MIL-STD-202.

Pull test: Test condition A, 10 pound pull.

Twist test: Test condition D.



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Solderability: Applicable.

Weight: 284 grams (10 ounces) maximum.

Identification marking (full): Applicable.

Life test requirements:

High level: 100,000 cycles minimum.

Low level: 100,000 cycles minimum plus 400,000 mechanical life.

Qualification inspection:

Qualification inspection and sample size: See table I.

TABLE I. Qualification inspection and sample size.

| Single submission | Group submission | |
|---|---|---|
| 18 units plus 1 unsealed unit for level L at C = 0 1/ | M28894/15-2320<br>M28894/15-5480 | 18 units plus 1 unsealed unit for level L at C = 0 1/ |
| 34 units plus 1 unsealed unit for level M at C = 0 1/ | | 34 units plus 1 unsealed unit for level M at C = 0 1/ |
| Qualification inspection as applicable. | | Qualification inspection as applicable. |
| | M28894/15-1360<br>M28894/15-4380<br>M28894/15-3400<br>M28894/15-3420 | 2 units each part number Qualification inspection, group 1. |

1/ The number of units for qualification testing will be increased as required in group V of the qualification inspection table of MIL-R-28894 if the contractor elects to test the number of units permitting one or more failures.

Part number: Consists of the basic number of this specification sheet and a dash number denoting mounting configuration, frequency trip point, and failure rate level.

| Example: | M28894/15- | 1 | 380 | L |
|---|---|---|---|---|
| Basic specification number. | | | | |
| Mounting configuration and frequency designator matrix (see figure 1 and table II). | | | | |
| Frequency trip point. Values may range from 320 to 480 Hz. Specify trip point using three digits. Example, for a trip point of 380 Hz, specify 380. | | | | |
| Failure rate level (see table III). | | | | |



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TABLE II. Mounting configuration and frequency type designator matrix.

| | Mounting style | | |
|---|---|---|---|
| | A | B | C |
| Under frequency | 1 | 2 | 3 |
| Over frequency | 4 | 5 | 6 |

TABLE III. Codes for failure rate level.

| Code letter | Failure rate level (% per 10,000 cycles) |
|---|---|
| L | 3.0 |
| M | 1.0 |
| P | 0.1 |
| R | 0.01 |

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AV
Navy - OS, SH

Preparing activity:
Navy - EC

Agent:
DLA - ES

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